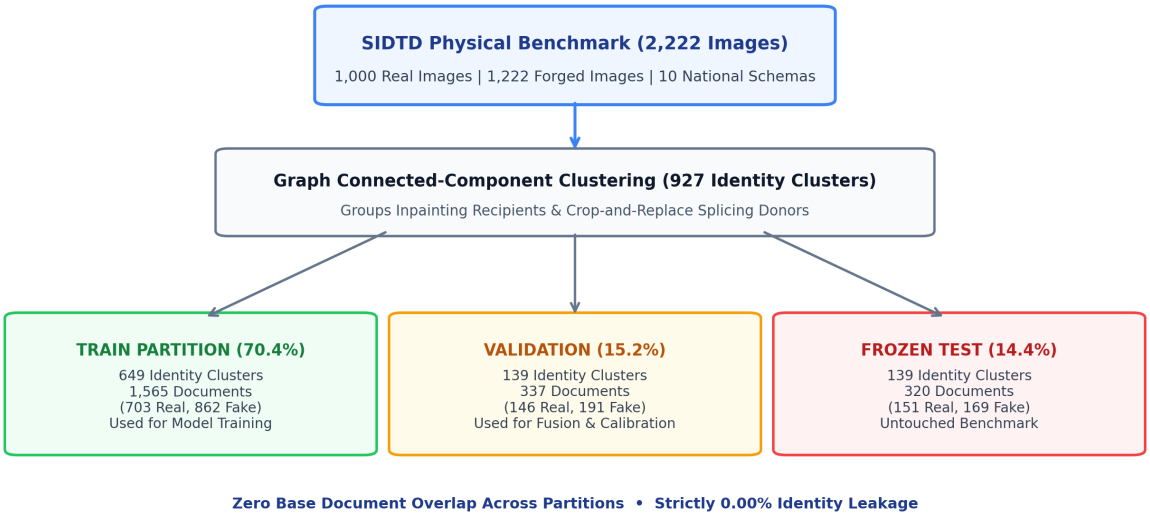
DocVerify — Phase 6: Oracle Leakage Audit & Frozen Test Manifest

Official Technical Milestone Report & Engineering Specification (Phase 6)

1. Phase Executive Summary & Objectives

Phase 6 conducted a thorough scientific integrity audit. The audit revealed a severe vulnerability: filename parsing (e.g. reading 'alb_id' from 'alb_id_01.jpg') acted as an oracle shortcut, bypassing true visual layout recognition. Phase 6 sealed this leakage, implemented autonomous geometry matching, and permanently froze the 320-sample test set.

2. Architectural Diagram & Pipeline Workflow



Zero Base Document Overlap Across Partitions • Strictly 0.00% Identity Leakage

3. Technical Specifications & Milestone Deliverables

Dimension / Metric	Implementation / Specification	Significance & Impact
Oracle Leakage Audit	Filename Dependency Identification	Revealed that the system crashed or cheated if images were renamed to 'scan_01.jpg'.
Autonomous Matcher	Geometry & Aspect Ratio Matching (`src/template_matcher.py`)	Measures physical width/height aspect ratios and layout features with zero reliance on filename hints.
UNKNOWN Fallback	Explicit UNKNOWN Template Routing	Anomalous dimensions or non-ID images are routed cleanly to global texture analysis without crashing.
Frozen Test Set	Permanent 320-Sample Benchmark Manifest	Locked frozen_test_manifest.csv (151 genuine, 169 forged) to ensure reproducible benchmarks.

4. Key Discoveries, Challenges & Next-Phase Directives

- Completely eliminated filename hints; achieved true blind document verification.
- Established permanent frozen test split ensuring all future benchmarks are 100% comparable.
- Sealed the project against shortcut learning and publication integrity criticisms.

DocVerify Technical Archive • Phase 6 Report Completed